

Why and Where to Use Kubernetes

Kubernetes is an open source container orchestration platform that automates the deployment, scaling, and management of containerised applications. Its benefits are manifold.



In software development, Kubernetes stands as a game-changer, redefining how applications are deployed and managed. As organisations adopt cloud-native approaches, understanding the significance of Kubernetes becomes essential for all stakeholders.

What is Kubernetes?

Kubernetes, often abbreviated as K8s, is an open source container orchestration platform designed to automate the deployment, scaling, and management of containerised applications. Originally developed by Google, Kubernetes is now widely adopted and maintained by the Cloud Native Computing Foundation (CNCF). It provides a robust and flexible framework for orchestrating containers, enabling seamless management of applications in various environments.

Key features of Kubernetes

Container orchestration: Kubernetes excels at managing containerised applications, enabling developers to deploy and scale containers seamlessly. It automates tasks such as load balancing, self-healing, and rolling updates, simplifying the operational aspects of containerised environments.

Scalability: Kubernetes supports automatic scaling of applications based on resource utilisation or custom metrics. This ensures that applications can handle varying workloads efficiently by dynamically adjusting the number of running containers.

High availability: Kubernetes enhances the reliability of applications by distributing containers across multiple nodes in a cluster. This ensures that if one node fails, the application can continue running on other nodes, providing high availability.

Portability: Kubernetes promotes a consistent deployment and management experience across various infrastructure providers, whether on-premise, in the cloud, or hybrid environments. This portability simplifies the migration of applications between different platforms.

Declarative configuration: Users define the desired state of their applications and infrastructure using YAML or JSON files. Kubernetes then continuously works to ensure that the current state matches the declared state, simplifying configuration management.

Core components of Kubernetes

The architecture of Kubernetes is designed to be highly modular and scalable. Here are the main components divided into logical parts.